

Lecture 3

In Class Exercises



Exercise 1

- Indicate, for each pair of expressions (A, B) in the table below whether A is Θ , Ω or O of B. Assume that $k \geq 1$, $\varepsilon > 0$, and $c > 1$ are constants. Write your answer in the form of the table with “yes” or “no” written in each box.

	A	B	O	Ω	Θ
a.	$\lg^k n$	n^ε			
b.	n^k	c^n			
c.	$\sqrt{n}$	$n^{\sin n}$			
d.	2^n	$2^{n/2}$			
e.	$n^{\lg c}$	$c^{\lg n}$			
f.	$\lg(n!)$	$\lg(n^n)$			

Exercise 1 (solution)

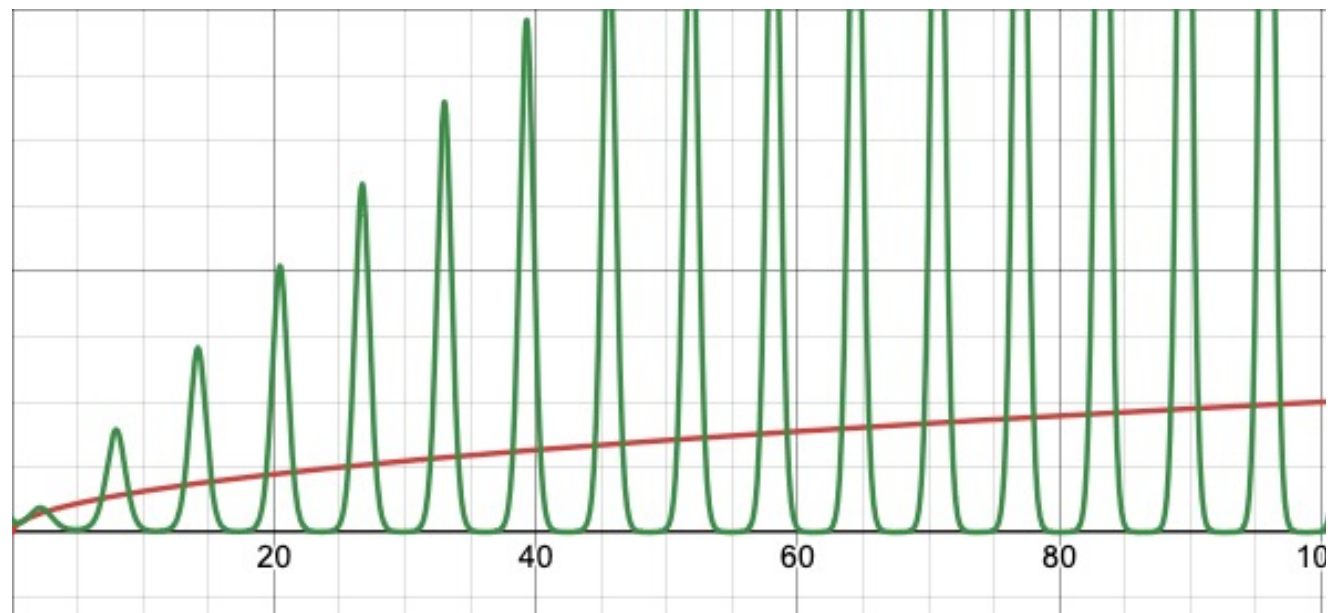
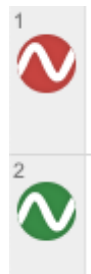
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	A	B	O	Ω	Θ
a.	$\lg^k n$	n^ε	yes	no	no
b.	n^k	c^n	yes	no	no
c.	$\sqrt{n}$	$n^{\sin n}$	no	no	no
d.	2^n	$2^{n/2}$	no	yes	no
e.	$n^{\lg c}$	$c^{\lg n}$	yes	yes	yes
f.	$\lg(n!)$	$\lg(n^n)$	yes	yes	yes

Exercise 1 (solution)

- ❑ a. Any polylogarithmic function is O of any polynomial function with a positive exponent.
- ❑ b. Any polynomial function is O of any exponential function with a positive base.
- ❑ c. The function $\sin n$ oscillates between -1 and 1. There is no value n_0 such that $\sin n$ is less than, greater than, or equal to $1/2$ for all $n \geq n_0$, and so there is no value n_0 , such that $n^{\sin n}$ is less than, greater than, or equal to $cn^{1/2}$ for all $n \geq n_0$.
- ❑ d. Take the limit of the quotient: $\lim_{n \rightarrow \infty} 2^n / 2^{n/2} = \infty$.
- ❑ e. These quantities are equal.
- ❑ f. $\lg(n!) = \Theta(n \lg(n))$. These functions are Θ of each other.

$$x^{\frac{1}{2}} \text{ vs } x^{\sin x}$$



Exercise 2

Is $2^{n+1} = O(2^n)$? Is $2^{2n} = O(2^n)$?

- Supplement your answers by providing a proof according to the O theorem (e.g. find a constant $c \dots$).

Exercise 2 (solution)

$2^{n+1} = O(2^n)$, but $2^{2n} \neq O(2^n)$.

To show that $2^{n+1} = O(2^n)$, we must find constants $c, n_0 > 0$ such that

$0 \leq 2^{n+1} \leq c \cdot 2^n$ for all $n \geq n_0$.

Since $2^{n+1} = 2 \cdot 2^n$ for all n , we can satisfy the definition with $c = 2$ and $n_0 = 1$.

To show that $2^{2n} \neq O(2^n)$, assume there exist constants $c, n_0 > 0$ such that

$0 \leq 2^{2n} \leq c \cdot 2^n$ for all $n \geq n_0$.

Then $2^{2n} = 2^n \cdot 2^n \leq c \cdot 2^n \Rightarrow 2^n \leq c$. But no constant is greater than all 2^n , and so the assumption leads to a contradiction.

Exercise 3

□ Prove the following theorem:

Theorem 3.1

For any two functions $f(n)$ and $g(n)$, we have $f(n) = \Theta(g(n))$ if and only if $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$. ■

Exercise 3 (solution)

If $f(n) = \Omega(g(n))$ then there exist $c_1 > 0$ and n_1 such that $f(n) \geq c_1 g(n)$ for all $n > n_1$. Furthermore, if $f(n) = O(g(n))$ then there exist $c_2 > 0$ and n_2 such that $f(n) \leq c_2 g(n)$ for all $n > n_2$. Therefore, if we set $n_0 = \max\{n_1, n_2\}$, then for all $n > n_0$, we have $c_1 g(n) \leq f(n) \leq c_2 g(n)$, which shows that $f(n) = \Theta(g(n))$.

The other direction is even simpler. Suppose $f(n) = \Theta(g(n))$. Then there exist $c_1, c_2 > 0$ and n_0 such that $c_1 g(n) \leq f(n) \leq c_2 g(n)$, which immediately shows that $f(n) = \Omega(g(n))$, as well as $f(n) = O(g(n))$.

Proof by induction (Recap)

- There are $\lg n + 1$ levels to a recursion tree for an input of size n .
- Base case: $n = 1 \Rightarrow 1$ level, and $\lg 1 + 1 = 0 + 1 = 1$.
- Inductive hypothesis is that a tree for a problem size of 2^i has $\lg 2^i + 1 = i + 1$ levels.
- Because we assume that the problem size is a power of 2, the next problem size up after 2^i is 2^{i+1} .
- A tree for a problem size of 2^{i+1} has one more level than the size- 2^i tree $\Rightarrow i + 2$ levels.
- Since $\lg 2^{i+1} + 1 = i + 2$, we are done with the inductive argument.

Exercise 4

- Use mathematical induction to show that when $n \geq 2$ is an exact power of 2, the solution of the recurrence

$$T(n) = \begin{cases} 2 & \text{if } n = 2, \\ 2T(n/2) + n & \text{if } n > 2 \end{cases}$$

- is $T(n) = n \lg n$.

Exercise 4 (solution)

□ Base case is when $n=2$: $n \lg n = 2 \lg 2 = 2 \cdot 1 = 2$.

□ For the inductive step, use the hypothesis:
 $T(n/2) = (n/2) \lg(n/2)$

□ Then

$$\begin{aligned} T(n) &= 2T(n/2) + n \\ &= 2(n/2) \lg(n/2) + n \\ &= n(\lg n - 1) + n \\ &= n \lg n - n + n \\ &= n \lg n \end{aligned}$$